MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG1DSCSDV100 – FUNDAMENTALS OF ACOUSTICS

Duration: 2 hrs. Maximum Marks: 70

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Part A

Very Short Answer Question

Answer **10** Questions

Each question carries **2** marks

1. What are partials in sound? [U] [CO2]
2. What defines a harmonic sound? [U] [CO2]
3. Identify one factor influencing sound reflection. [K] [CO4]
4. How is sound refraction defined? [U] [CO4]
5. What unit is used to measure acoustic impedance? [K] [CO5]
6. What kinds of materials are effective at absorbing sound? [U] [CO5]
7. What is the role of the outer ear? [U] [CO6]
8. What is the meaning of SIL? [K] [CO6]
9. How would you define sound transmission? [U] [CO3]
10. What does medium density refer to? [K] [CO3]
11. How would you define sound? [U] [CO1]
12. what is the unit of frequency? [K] [CO1]

[2 x 10 = 20]

Part **B**

Short Answer Type Questions

Answer **6** Questions

Each question carries **5** marks

13. How does temperature affect sound transmission in gases, and why? [U] [CO3]
14. Explain how midrange frequencies contribute to sound clarity in environments like theaters or classrooms. [A] [CO6]
15. Explain the role of the cochlea in hearing. [U] [CO6]
16. How you can sound proof a room? [A] [CO5]
17. Explain the Role of bass traps in acoustic design [U] [CO6]
18. Expalin is law of reflection ? [U] [CO4]
19. How does the fundamental frequency impact pitch perception in complex sounds [U] [CO2]

20. How do overtones contribute to the perception of timbre in different musical instruments? [U] [CO2]
21. What is phase cancellation? Explain. [U] [CO1]

[5 x 6 = 30]

Part **C**
Essay Type Questions
Answer **2** Questions
Each question carries **10** marks

22. what is acoustic imedance ? Explain significans and facorts affecting Acostic impedance? [U] [CO5]
23. Analyze the relationship between the density of a medium and the speed of sound. Provide examples of how sound travels differently in materials with varying densities. [An] [CO3]
24. Explain the Characteristics of sound and sound wave [U] [CO1]

[10 x 2 = 20]